INTERNSHIP LIFE-CYCLE TRACKER

Enterprise Portal & Cloud Database Integration Report

PREPARED FOR: CENTRAL UNIVERSITY OF TECHNOLOGY
DIRECTORATE: ACADEMIC AFFAIRS & PLACEMENT CELL
DATE: AUGUST 1, 2026
VERSION: 1.0.0 (STABLE PRODUCTION)
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1. Executive Summary

The Internship Life-Cycle Tracker (InternFlow Portal) is an enterprise web application designed to manage the end-to-end student industry placement workflow. The system eliminates fragmented manual processing by automating application approvals, daily journal logging, document vault checks, HOD and Mentor clearances, and placement-driven No Objection Certificate (NOC) generation. By integrating Supabase for cloud data storage and Google Gemini AI for contract auditing, the portal secures, automates, and validates university academic internships.

2. System Architecture & Role Permissions

The application utilizes a multi-tier client-server structure. The React 18 client interfaces with a Node.js/Express middle-tier REST API, which connects directly to a hosted Supabase Postgres database. RBAC (Role-Based Access Control) gates all portal features:

- **Student**: Log application forms, submit resumes and contract files, and log daily logs.
- **Academic Mentor**: Validates student offer details and weekly journals.
- **HOD (Head of Department)**: Reviews department placement analytics and provides departmental clearances.
- **Placement Cell Officer**: Endorses student records, manages corporate directory partners, and signs off digitally generated NOCs.
- **System Admin**: Reviews active threat logs, adds/removes accounts, and tracks configurations.

3. Business Rules & Functional Features

- **File Deduplication**: Uploading duplicate items replaces existing slots in-place, maintaining document vault integrity.
- **Mandatory Credentials**: Student submissions require both Resume and Offer Letter to proceed.
- **Role Constraints**: Document upload operations are restricted solely to Students, preventing staff record tempering.
- **NOC Printing**: Dynamic HTML-to-PDF print popups allow direct local PDF exports when signed by placement cells.
- **Offline Fail-Safe**: Backend fallbacks have been removed; system failures report 500 error flags to ensure direct cloud state validation.

4. Database & Storage Models

The relational Postgres tables in Supabase manage structured entities, while Supabase Storage handles binary PDF/image resources using Row-Level Security (RLS) policies.

Table Name	Primary Purpose	Security Model
profiles	Portal users, titles, and roles	RLS Read-All (Public)
applications	Internship details and workflow states	RLS Read-All (Public)
student_documents	Relation table linking files to storage	RLS Read-All (Public)
weekly_reports	Student weekly logbook and grading	RLS Read-All (Public)
partner_companies	Directory of verified company partners	RLS Read-All (Public)
audit_logs	Admin security audit log registry	RLS Read-All (Public)

5. AI Document Verification & Compliance

The portal is connected to the Google Gemini API. When a student uploads an Offer Letter, the system scans the text structure, extracts crucial values (e.g. stipend, location, duration), and evaluates policy compliance (minimum 16 weeks duration, prior approval requirement). A verification score is calculated and warning flags are logged to support academic mentor decisions.

6. Project Implementation & Verification Summary

The system has been successfully verified, building with 0 compilation errors across both frontend and backend TypeScript configs. All components (steppers, grids, tables, print modals, and RLS configurations) have been integrated and deployed.